

Formulating Queue Management as an MDP

CORE OBJECTIVE

Balance service speed against operational costs in a single-server queue. Dynamically adjust service rates to minimize cumulative expected holding and service costs.

State & Action Space

State Space S

$\{0, 1, 2, \dots, 15\}$
 $s = \# \text{ customers in queue}$

Action Space A

0: Normal Service
1: Fast Service

Dynamics & Rewards

Parameters: $\lambda = 0.3$ | $\mu_0 = 0.4$ (Normal) | $\mu_1 = 0.7$ (Fast)

Reward Function $R(s, a)$

$R(s, a) = -(2s + \text{service_cost})$
Holding = $2s$ | Normal = 1 | Fast = 5

Goal: Solve for optimal policy π with $\gamma = 0.95$:

$$\max_{\pi} E \left[\sum \gamma^t R(s_t, a_t) \right]$$

Results: Optimal Threshold Policy Validated

ALGORITHM EFFICIENCY

3 Iterations

CONVERGENCE SPEED

0.02 Seconds

TOTAL RUNTIME

Policy iteration demonstrates rapid convergence to a stable value function.

OPTIMAL POLICY

QUEUE LENGTH (S)	ACTION (A)
0 – 2	Normal Service
≥ 3	Fast Service

Threshold is reached when holding costs outweigh fast service premiums.

THEORETICAL INSIGHT

The results align with **Classical Queue Control Theory**, confirming a threshold-based structure.

Management Rule:

Prioritize cost during stability; prioritize speed during congestion.

The optimal strategy provides a mathematical foundation for dynamic resource allocation, ensuring system stability while minimizing operational expenditure.

KEY INSIGHT